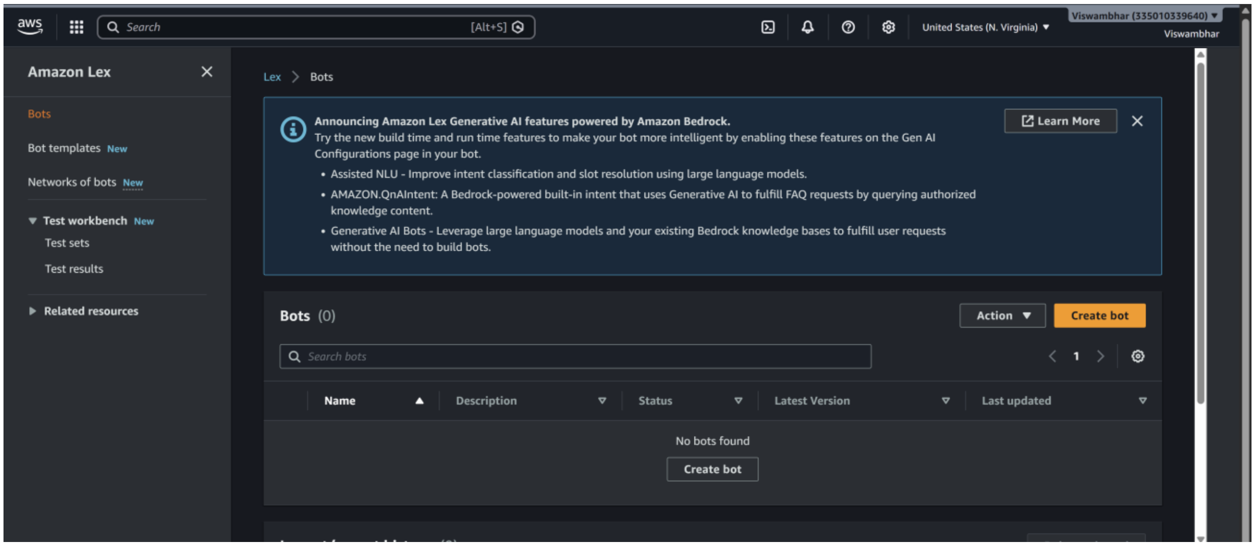
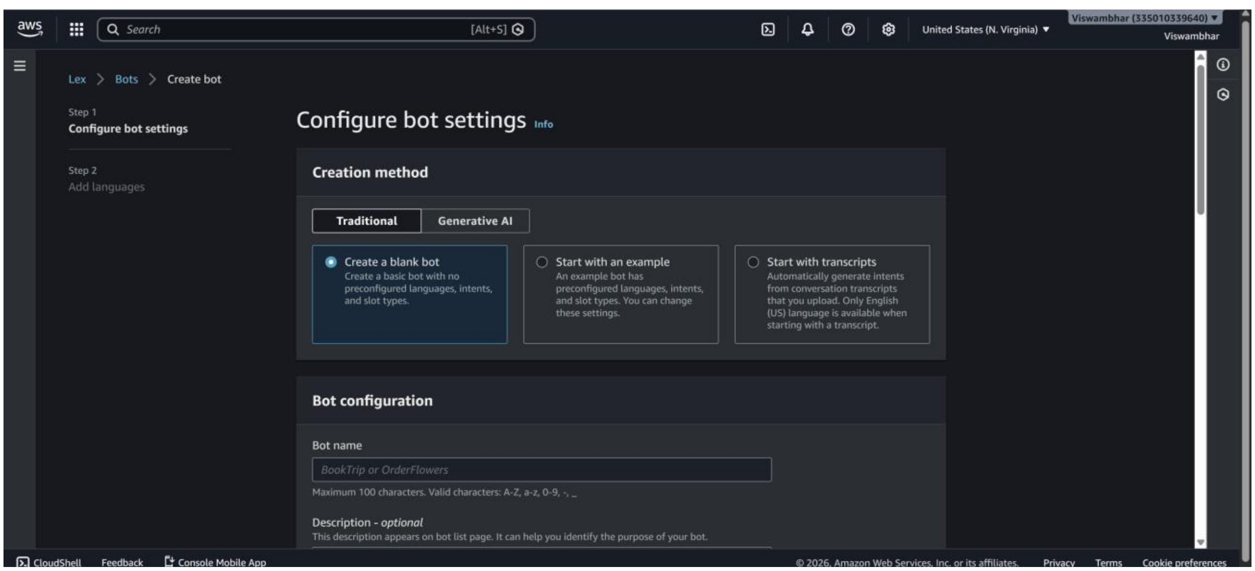
WEEK-11   Amazon LEX

Step 1: Open Amazon Lex

1. Login to AWS Console
2. Search **Amazon Lex**

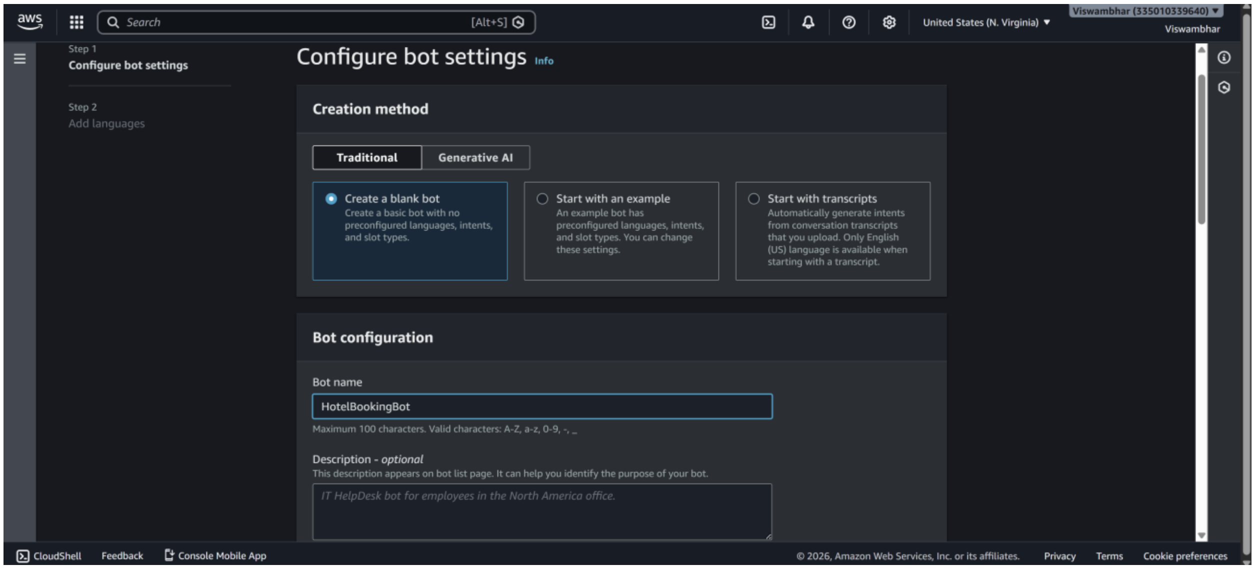


3. Click **Create bot**

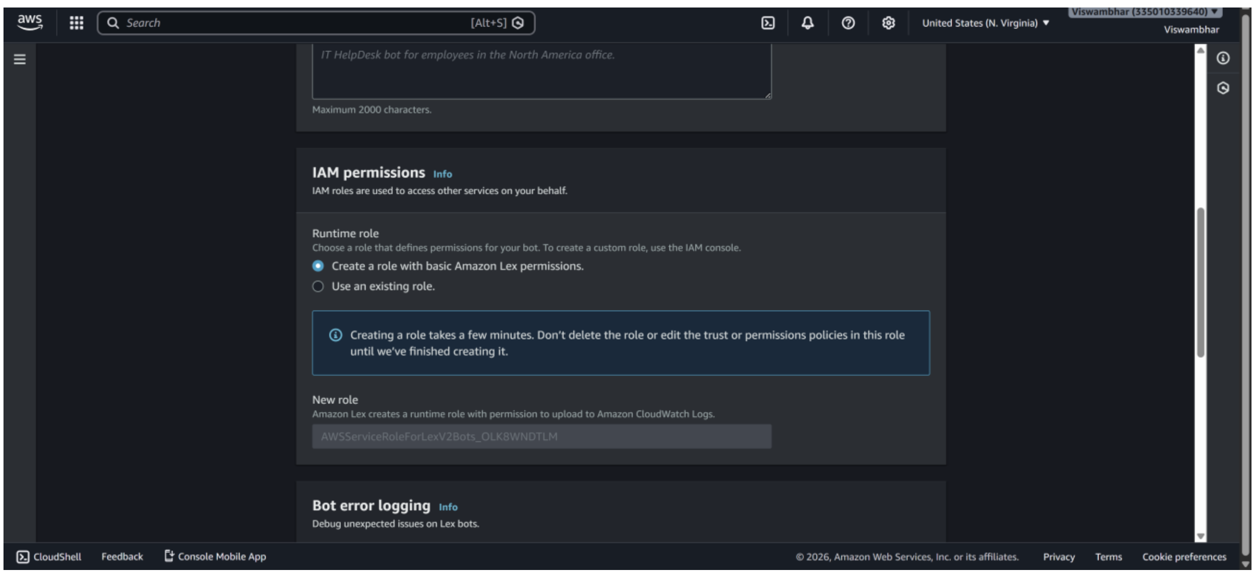


Step 2: Create Bot

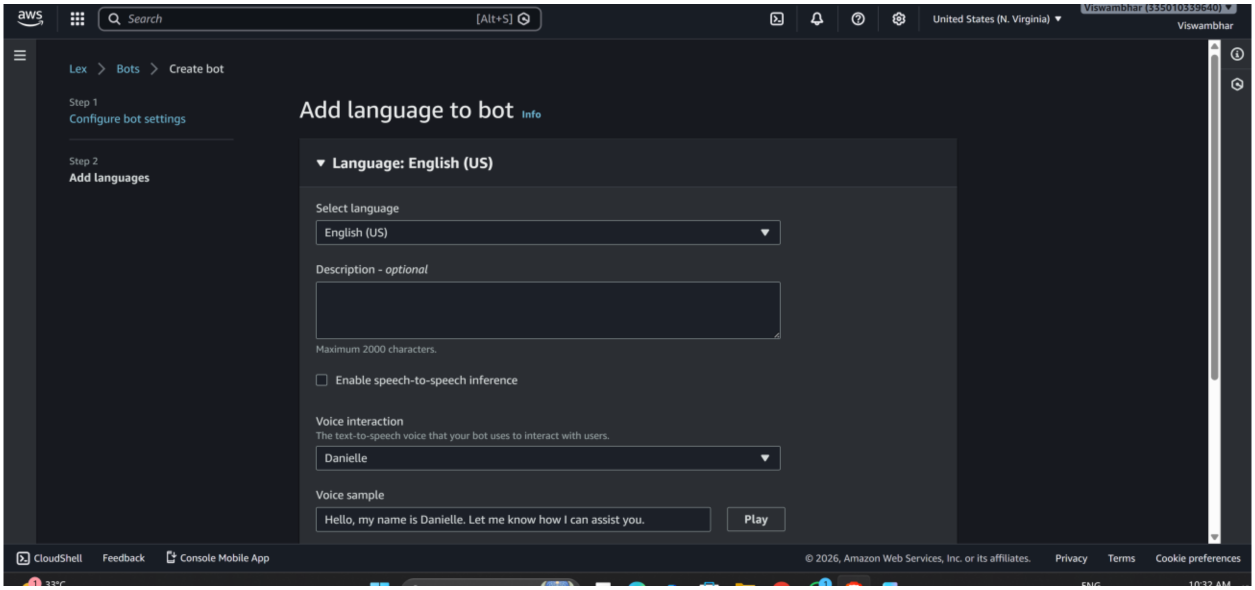
1. Choose:
    - **Create a blank bot**
2. Enter:
    - Bot name: HotelBookingBot



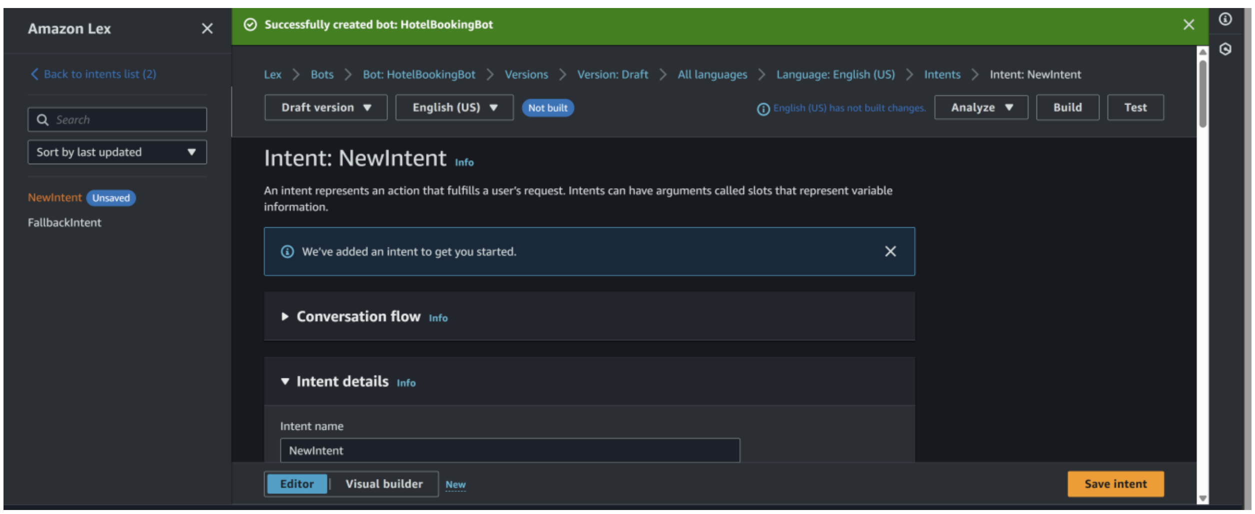
3. IAM role → Create new role



4. Remaining default→next
5. Language: English

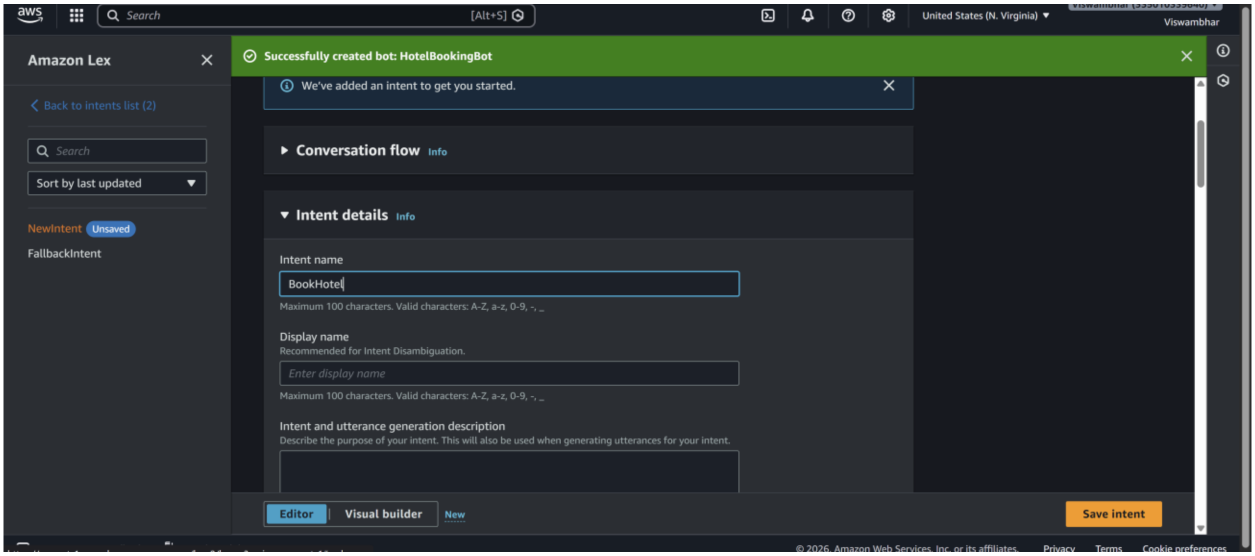


6. Voice (optional)
7. Click **Done**



Step 3: Create Intent

1. Go to **Intents**
2. Name: BookHotel



3. Click **Create**

Step 4: Add Sample Utterances

Add user sentences (how user talks):

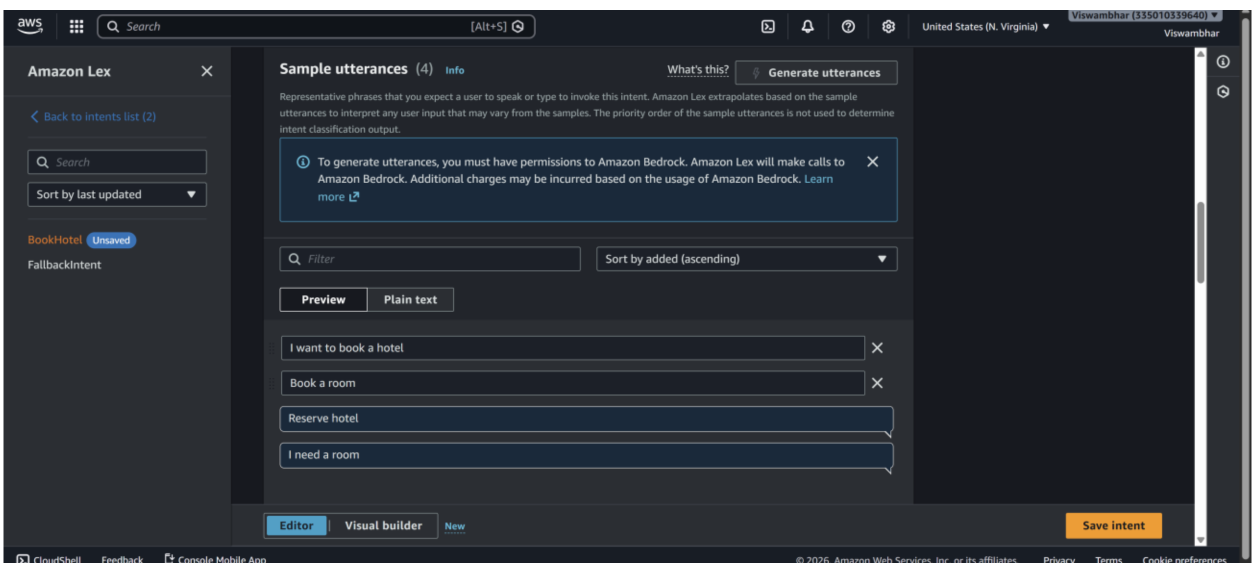
I want to book a hotel

Book a room

Reserve hotel

I need a room

These are **Utterances**



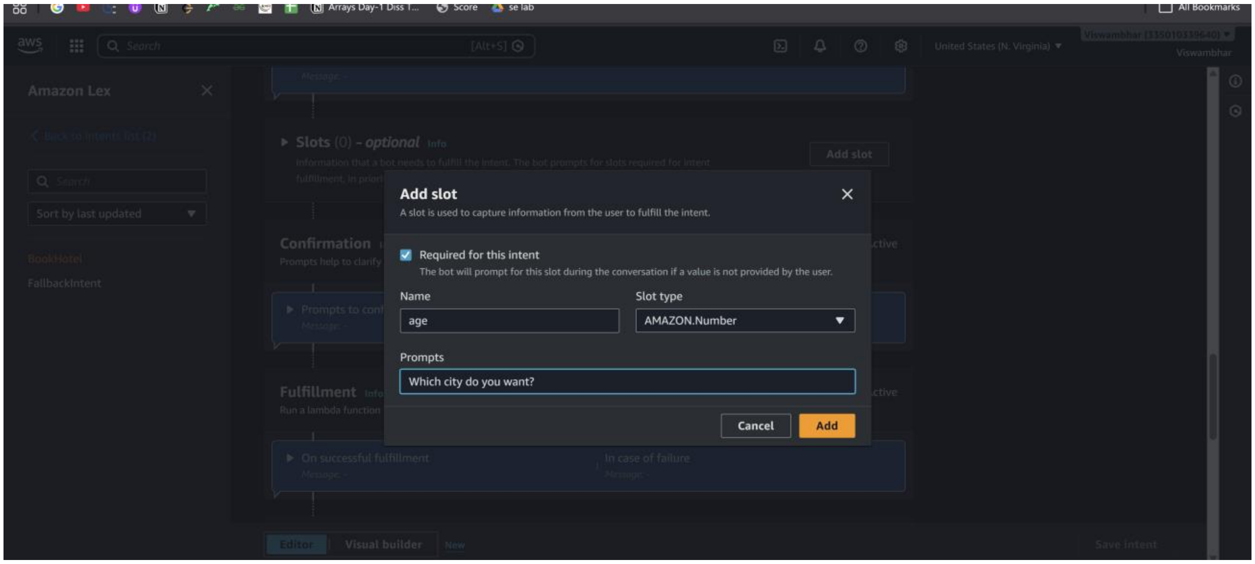
Step 5: Create Slots (User Inputs)

Slots collect user information.

Add Slot

1. Click **Add slot**
2. Fill:

- Slot name: age
- Slot type: AMAZON.Number
- Prompt: *Which city do you want?*



Mark as **Required**

Add IF Conditions (Conditional Logic)

Used to control flow based on slot values

Example Condition:

If {age} < 18

Steps:

1. Go to **Intent → Slots -> under Advance options**
2. Scroll to **Slot capture: success response ->Conditional branching**
3. Click **Add condition**

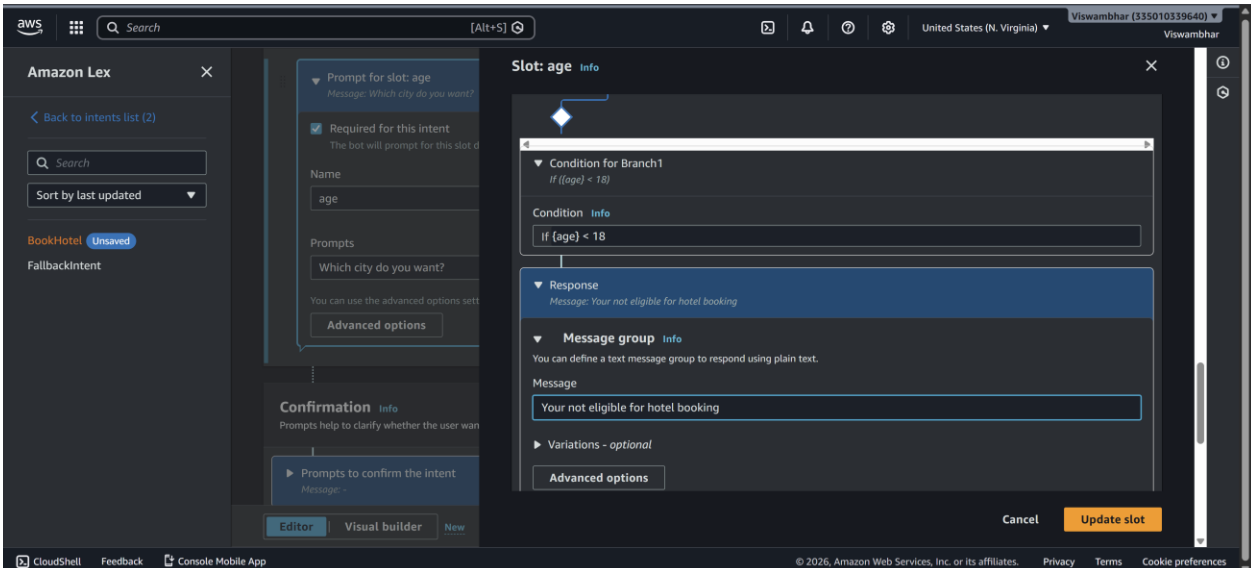
Example:

Condition:

If {age} < 18

Response:

Your not eligible for hoot booking?

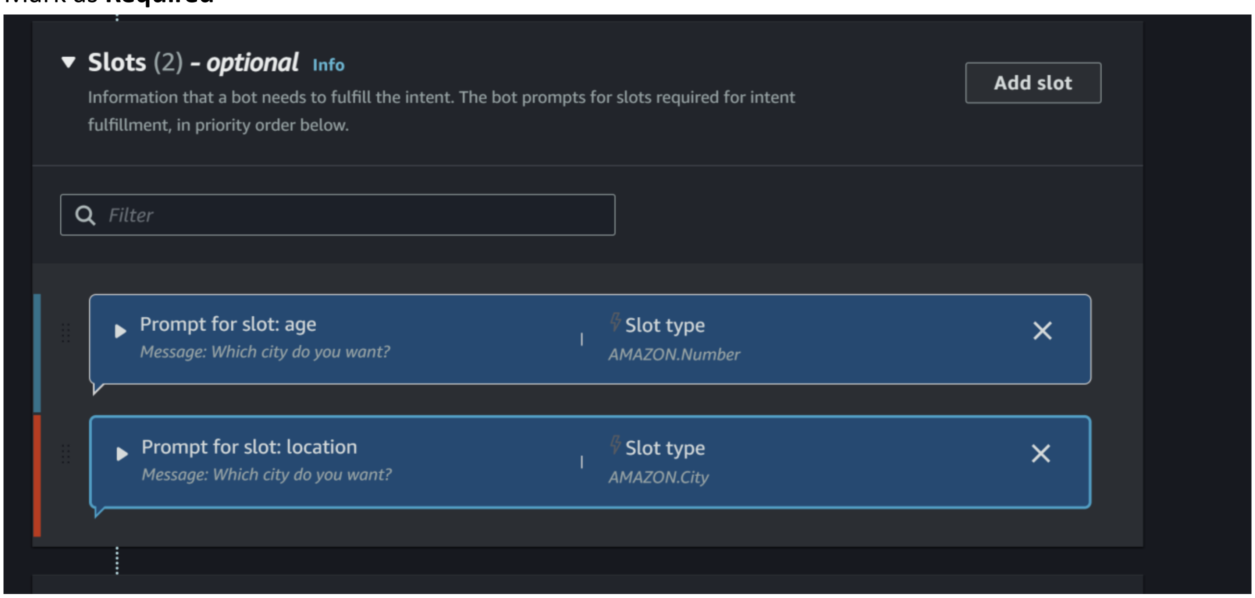


Step 6: Create Slots (User Inputs)

Slots collect user information.

Add Slot

3. Click **Add slot**
4. Fill:
   - Slot name: location
   - Slot type: AMAZON.City
   - Prompt: *Which city do you want?*
5. Mark as **Required**



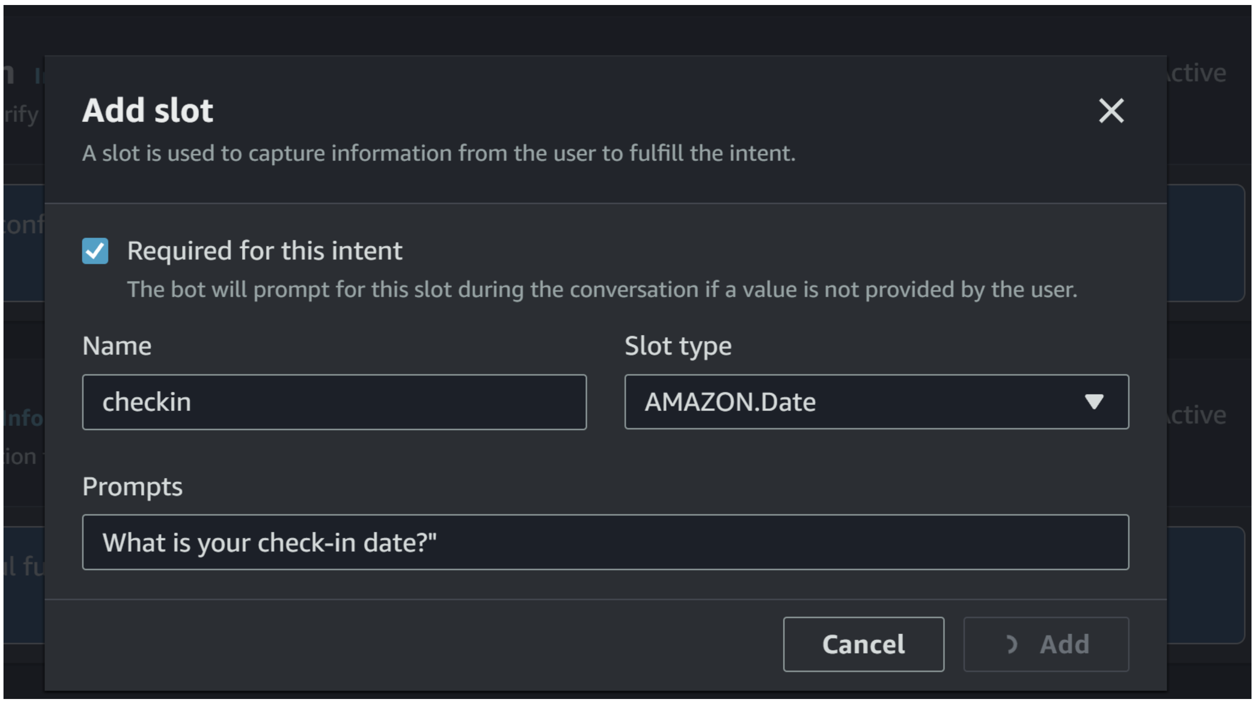
6.

Repeat for other slots.

1. Click **Add slot**
2. Fill details:

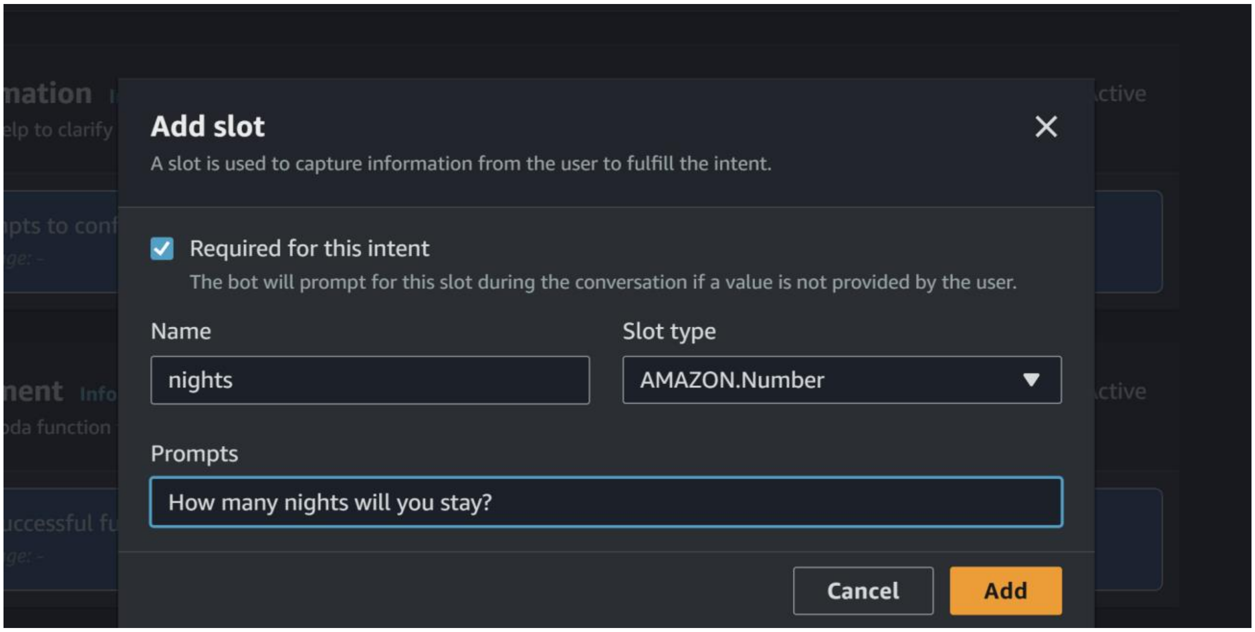
- **Slot name:** checkin
- **Slot type:** AMAZON.Date
- **Prompt:**
  "What is your check-in date?"

3. Mark as **Required**
   (Optional but recommended)
   - Add **Retry prompt**:
     "Please provide a valid date (e.g., 2026-03-25)"
4. Click **Save**



Slot 7: Number of Nights (nights)
Steps:

1. Click **Add slot** again
2. Fill details:
   - **Slot name:** nights
   - **Slot type:** AMAZON.Number
   - **Prompt:**
     "How many nights will you stay?"
3. Mark as **Required**
4. Click **Save**



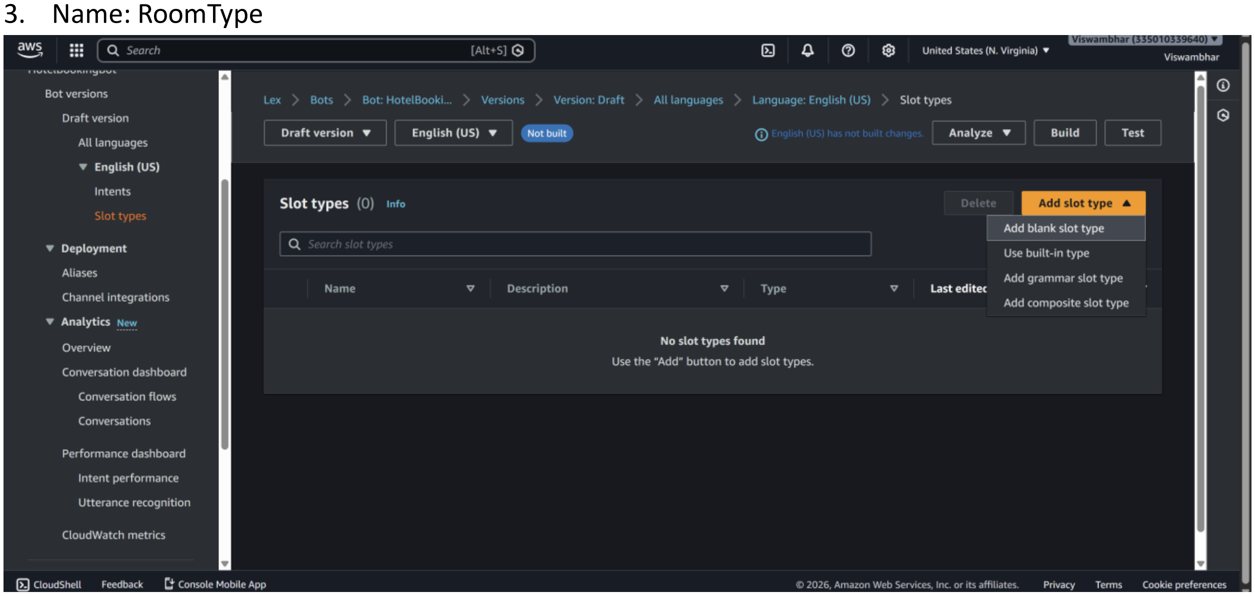
Step 7: Create Custom Slot Type

Save intent

Click on Back to intents

If you want custom values:

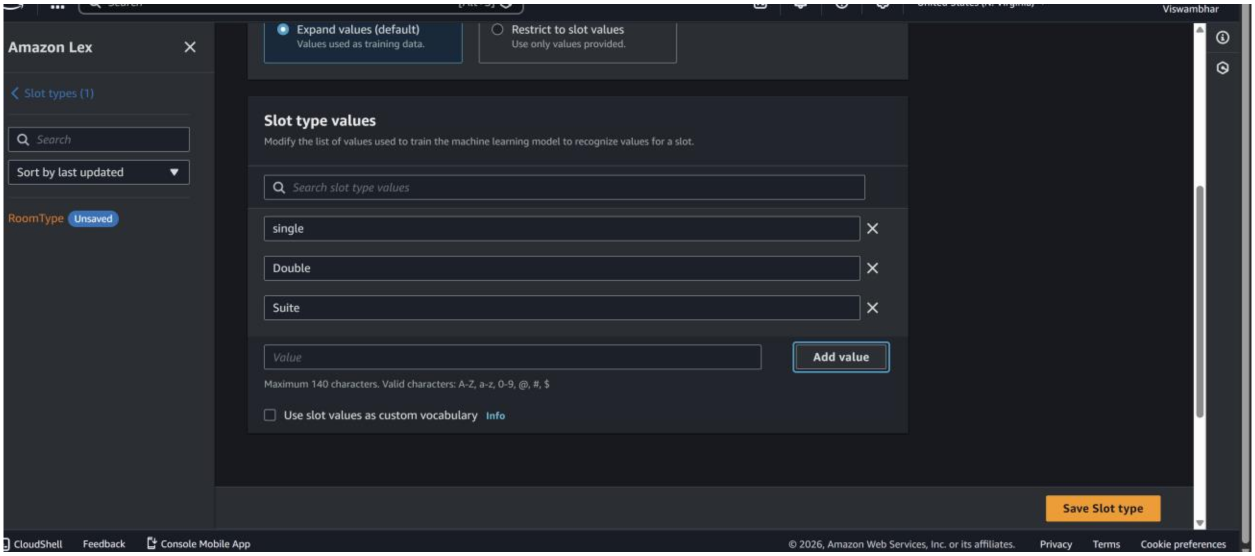
1. Go to **Slot Types**
2. Click **Add slot type ->add blank slot type**
3. Name: RoomType



Add values:

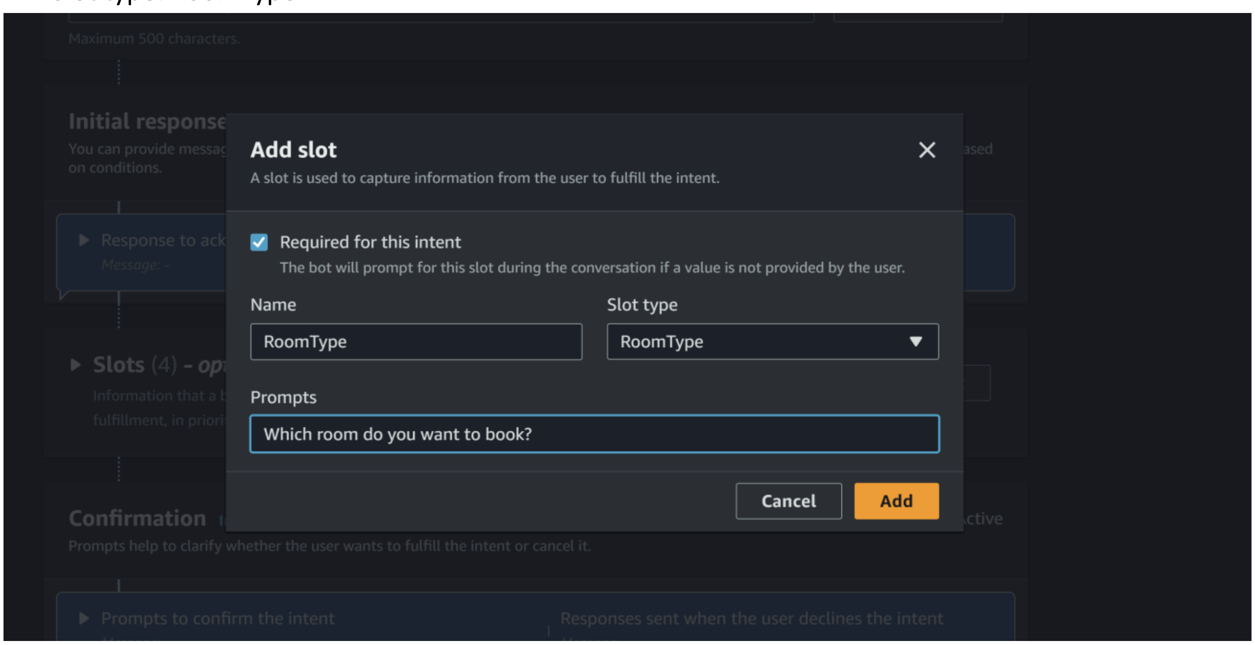
Single

Double
Suite
Save slot type



Now use this in slot: go back to intent

- Slot name: RoomType
- Slot type: RoomType

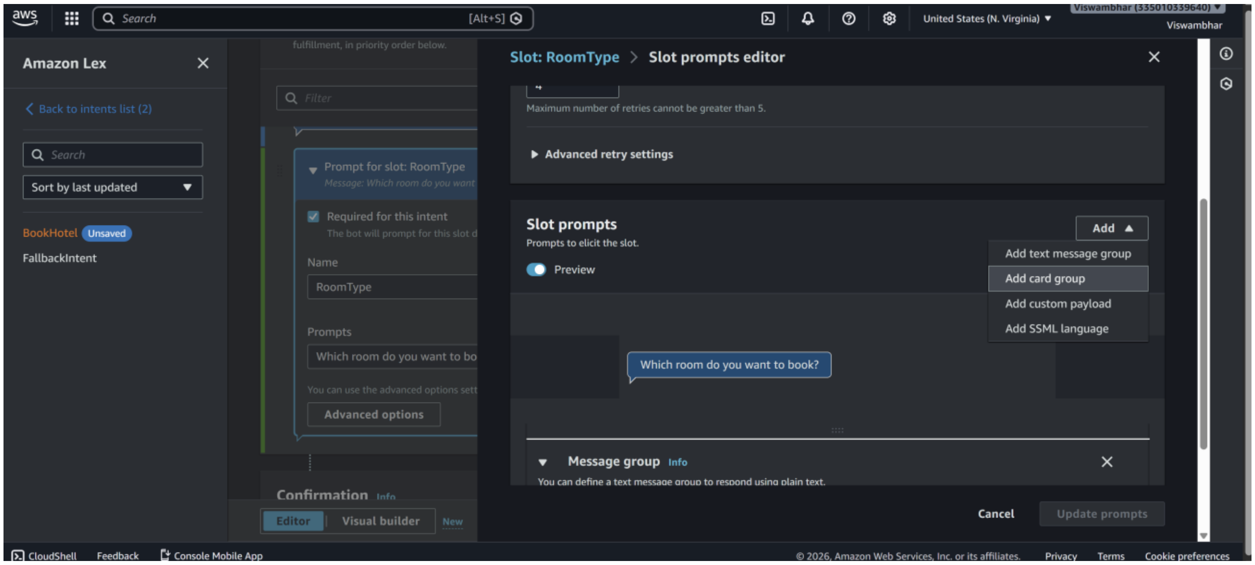


Step 9: Add Response Cards (Card Groups)

Used for buttons (quick replies)

Steps:

1. Go to **slot prompts -> more prompt options**
2. Click **Add -> add card groups**

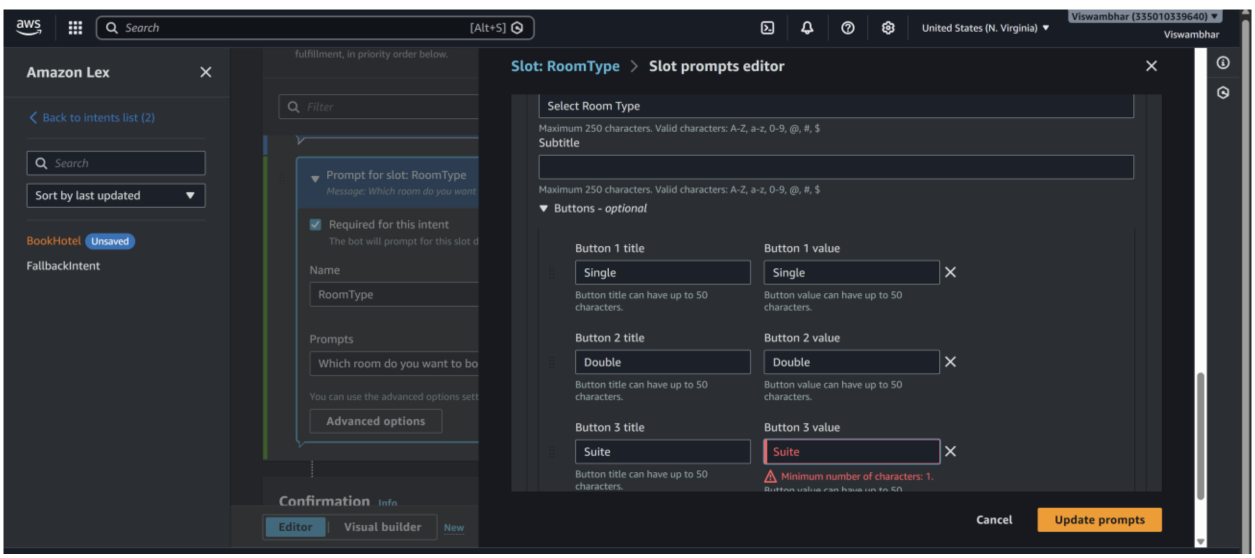


Example:

Card Title: Select Room Type

click on add Buttons:

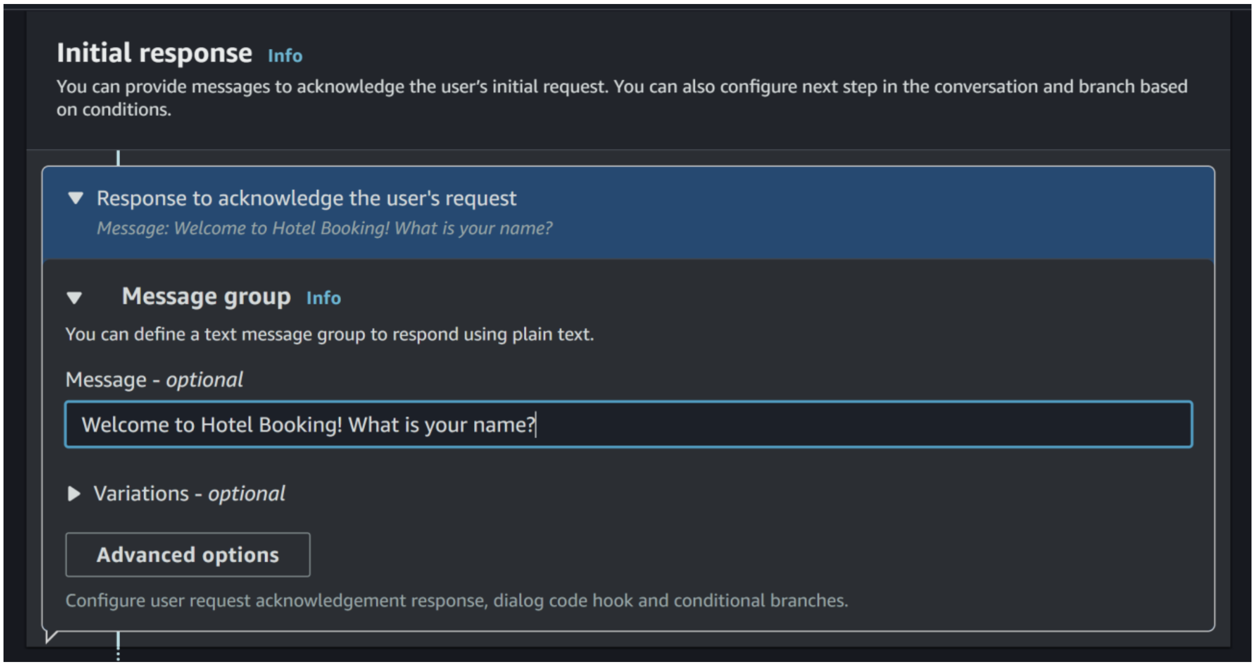
- Single → "Single"
- Double → "Double"
- Suite → "Suite"



Step 10: Configure Responses

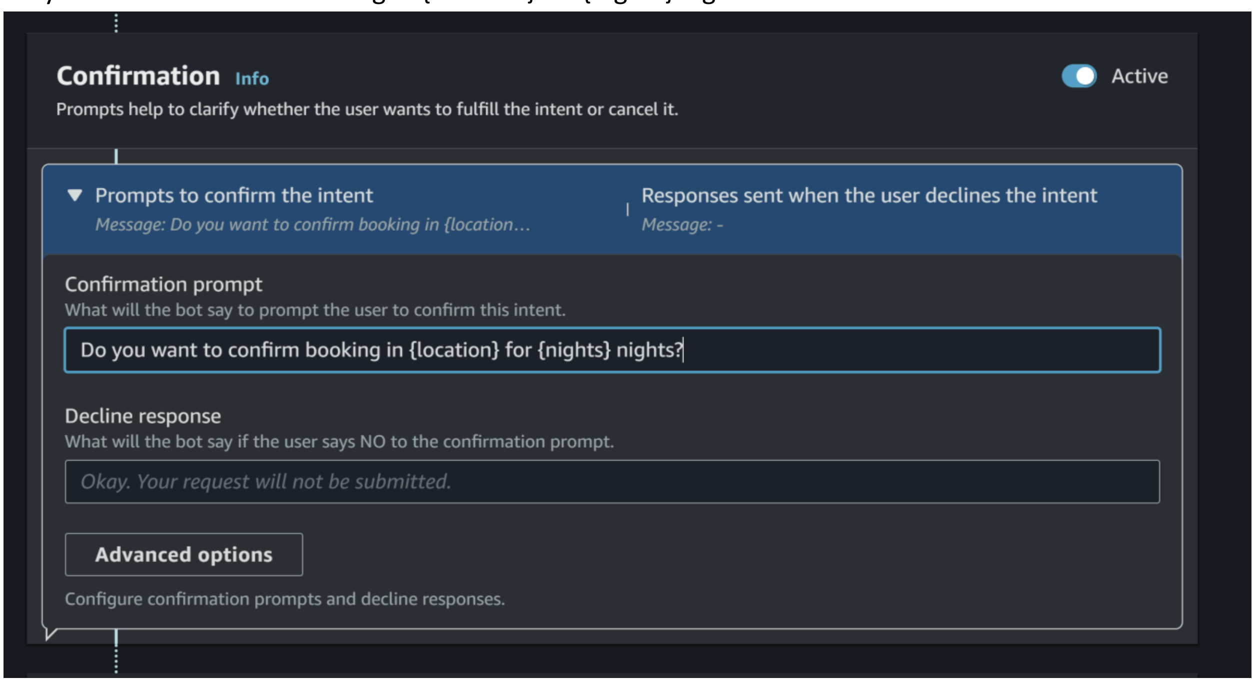
Initial Response:

Welcome to Hotel Booking! What is your name?



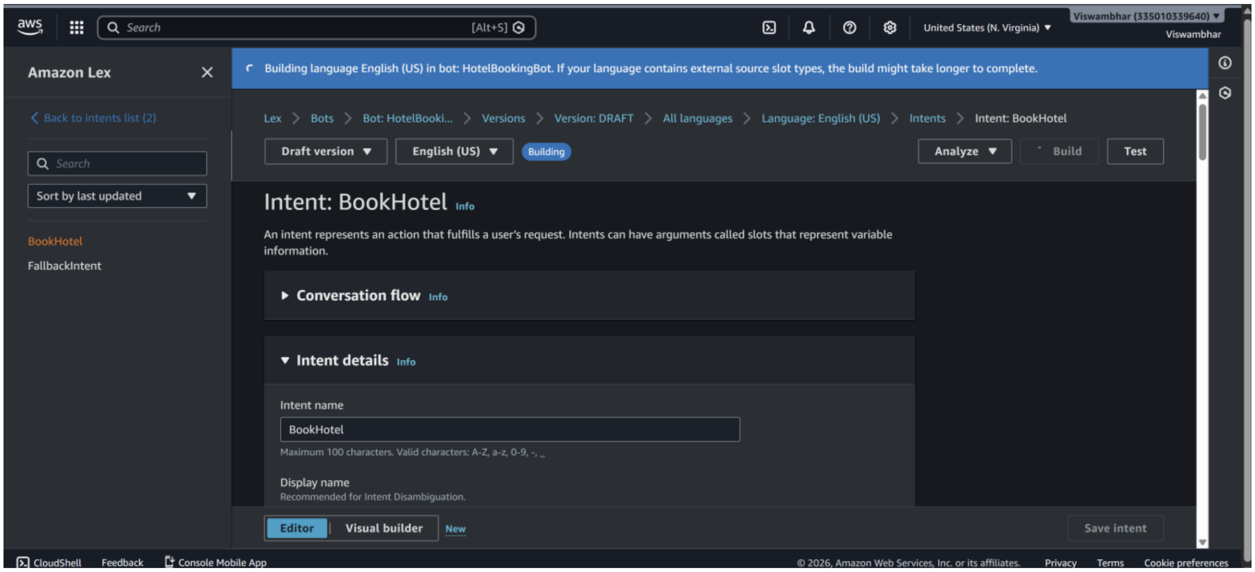
Confirmation:

Do you want to confirm booking in {location} for {nights} nights?



Step 11: Build & Test

1. Click **Build**
2. Use **Test chatbot** panel
3. Try:



Book a hotel

Sample output

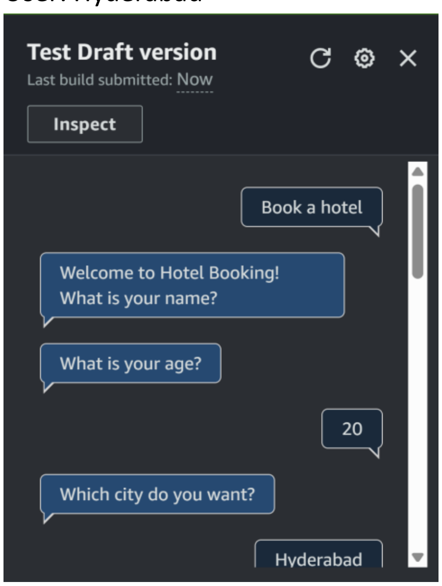
User: Book a hotel

Bot: What is your age?

User: 25

Bot: Which city do you want?

User: Hyderabad



Bot: Check-in date?

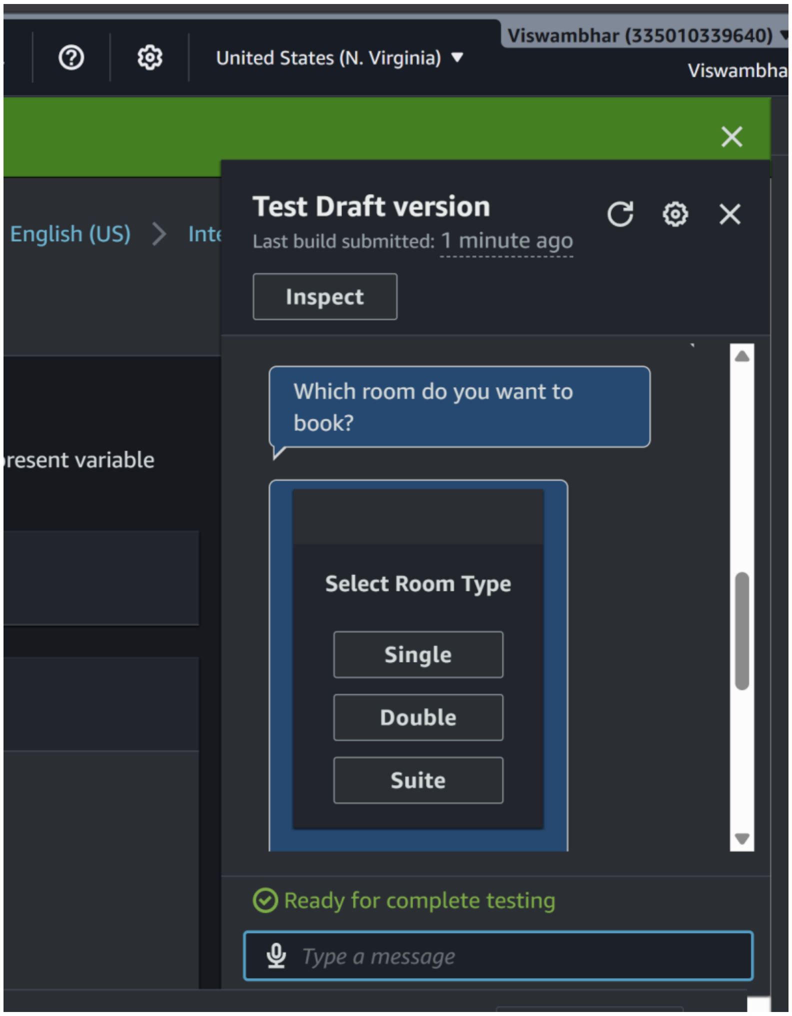
User: Tomorrow

Bot: Number of nights?

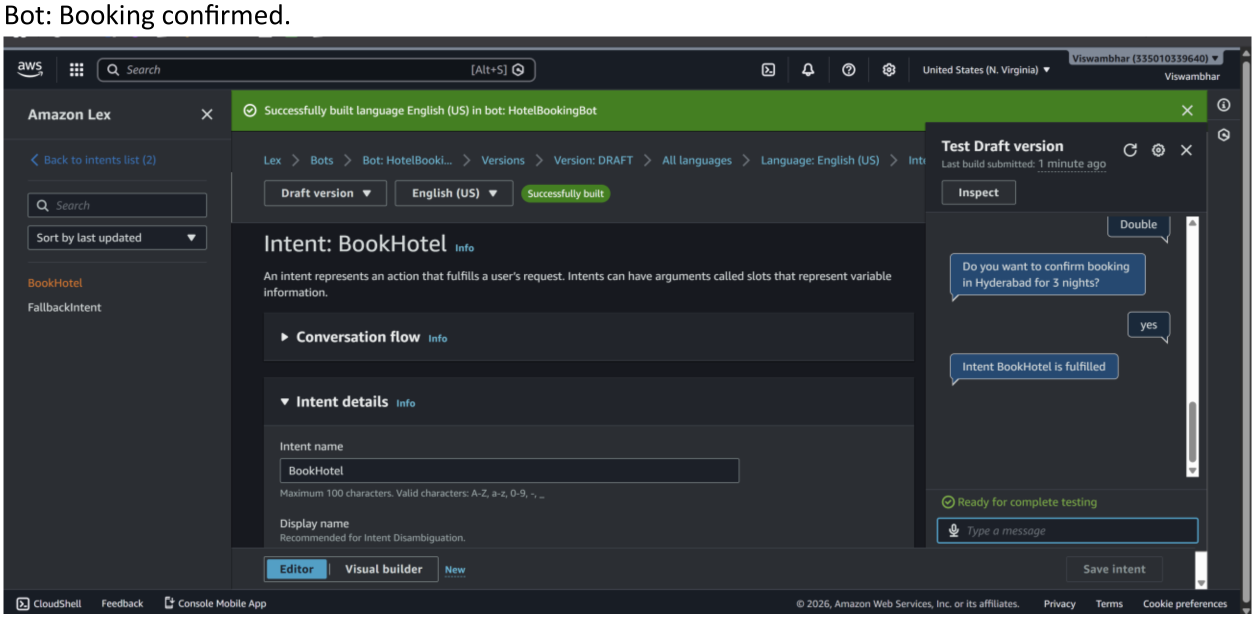
User: 2

Bot: Select room type

User: Double



Bot: Do you want to confirm booking in Hyderabad for 2 nights?
User: Yes
Bot: Booking confirmed.



SCENARIO BASED QUESTIONS

Scenario 1: Bot Not Responding

You created a chatbot, but when you test it, the bot does not respond to user input.

• What could be the possible reason for no response?

- Bot not built
- No matching intent
- Wrong version/alias
- Errors in configuration

• Did you build and publish the bot correctly?

- Bot must be built after changes
- Must publish version and assign alias

• How would you troubleshoot this issue?

- Rebuild the bot
- Check intents and utterances
- Test with simple inputs
- Check logs

Scenario 2: Incorrect Intent Matching

A user types a valid query, but the chatbot responds with the wrong intent.

• What could cause incorrect intent recognition?

- Poor or insufficient utterances
- Similar intents causing confusion

• How do utterances affect intent matching?

- They train the model
- More variations improve accuracy

• How can you improve accuracy?

- Add diverse utterances
- Reduce overlap between intents

Scenario 3: Missing Slot Values

The chatbot asks for user input, but fails to capture required slot values.

• What is a slot in Amazon Lex?

- A variable used to capture user input (e.g., city, date)

• Why might slot values not be captured properly?

- Incorrect slot type
- Poor prompt design
- Invalid user input

• How would you fix slot configuration issues?

- Match slot type with prompt
- Add retry prompts
- Validate input

Scenario 4: Fulfillment Not Working

The bot collects all user inputs but does not perform the expected action.

• What is fulfillment in Amazon Lex?

- The process of executing actions after collecting inputs

• Could Lambda integration be misconfigured?

- Yes, wrong permissions or configuration

• How would you verify the backend processing?

- Check Lambda logs
- Test function separately

Scenario 5: Bot Not Accessible

You created the bot successfully, but users are unable to access it.

• Did you deploy or publish the bot?

- Bot must be published

• Could permissions or channel integration be missing?

- Yes, IAM roles or integration may be missing

• How would you make the bot accessible to users?

- Publish bot
- Set alias
- Integrate with application

Scenario 6: Wrong Language Understanding

The chatbot fails to understand user queries written in slightly different formats.

• How does Amazon Lex process natural language?

- Uses NLP and machine learning

• Why is training with multiple utterances important?

- Improves understanding of variations

• How can you improve NLP performance?

- Add more utterances
- Use synonyms and variations

Scenario 7: Versioning Issue

You updated the bot, but users still experience the old behavior.

• Did you create a new version of the bot?

- Changes require a new version

• Why is alias configuration important?

- Alias points to a specific version

• How do you ensure users access the latest version?

- Update alias to latest version

Scenario 8: Lambda Permission Error

The chatbot is connected to a Lambda function, but it fails during execution.

• What type of permission error might occur?

- Access denied error

• How are IAM roles used in this integration?

- Control permissions between services

• How would you fix the permission issue?

- Update IAM policies
- Allow Lex to invoke Lambda

Scenario 9: Bot Gives Default Response

The chatbot always gives a fallback/default response instead of proper answers.

• What is a fallback intent?

- Default intent when no match is found

• Why does the bot fail to match user input?

- Poor utterances

- Weak training

• How can you improve intent recognition?

- Add more utterances
- Improve intent design

Scenario 10: Multi-turn Conversation Issue

The chatbot fails to continue conversation properly after asking follow-up questions.

• What is dialog management in Amazon Lex?

- Managing conversation flow using slots

• How are slots used in multi-turn conversations?

- Collect user inputs step-by-step

• How can conversation flow be improved?

- Proper slot order
- Clear prompts

Scenario 11: Integration with Web Application

A company wants to integrate the chatbot into their website.

• How can Amazon Lex be integrated with web applications?

- Using APIs or AWS SDK

• What services or APIs are required?

- Lex API, API Gateway, Lambda

• What challenges might occur during integration?

- Authentication
- Latency
- UI handling

Scenario 12: Performance Issue

The chatbot responds slowly when many users interact simultaneously.

• What could cause performance delays?

- High traffic
- Slow backend

• How can scalability be handled in Amazon Lex?

- Use serverless scaling

• What AWS services can support high traffic handling?

- Lambda
- API Gateway
- CloudWatch